Seyyed Ali Aghamiri

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Research Interests: Distributed, Networked, and ML Systems; Resource Scheduling & Optimization

EDUCATION

University of Tehran ***Sep 2023 - Present***
M.Sc. in Computer Engineering - Computer Architecture
- GPA: 17.15/20.00
- Supervisor: Prof. Ahmad Khonsari
- Thesis: Optimization of tip selection algorithm on DAG based blockchain in IOT.
- Relevant coursework: Advanced Computer Networking, Performance Evaluation of Computer Systems, Cryptocurrencies, Internet Measurement, Cyber-Physical Systems (CPS).

Persian Gulf University ***Sep 2017 - Aug 2022***
B.Sc. in Computer Engineering - Software Engineering
- GPA: 15.61/20.00

PUBLICATIONS

[J1] **Seyyed Ali Aghamiri**, A. Karimi, M. Dolati, and A. Khonsari, "Compute and IO-Aware Distributed Training Management in Converged Networks," *IEEE Transactions on Cloud Computing*, **submitted**.

[C2] M. Mozhganfar, P. Jamshidi, **Seyyed Ali Aghamiri**, M. Ghasemi, M. Dolati, F. Tashtarian, A. Khonsari, and C. Timmerer, "YTLive: A Dataset of Real-World YouTube Live Streaming Sessions," *IEEE/IFIP Network Operations and Management Symposium (NOMS)*, 2026.

[C1] **Seyyed Ali Aghamiri**, R. Sharifnia, and A. Khonsari, "Utilizing Transaction Prioritization to Enhance Confirmation Speed in the IOTA Network," 29th International Computer Conference, Computer Society of Iran (CSICC), 2025.

RESEARCH EXPERIENCE

Distributed Training and ML Systems ***2025 - Present***
University of Tehran; advised by Dr. Mahdi Dolati
- Developed and implemented the optimization and heuristic components of NEXUSTRAIN, a resource-aware multi-job distributed training scheduler that jointly considers hybrid data/pipeline parallelism, GPU placement, compute scheduling, remote-data prefetching, and shared network/storage constraints.
- Formulated the scheduling problem as a time-indexed mixed-integer linear program and developed scalable search and resource-aware scheduling procedures.
- Implemented the solver and heuristic framework, adapted the TopoOpt/FlexNetPacket-based simulation environment, and conducted numerical and packet-level evaluations.

IOTA / DAG-Based Distributed Ledgers ***2023 - Present***
University of Tehran; Supervisor: Prof. Ahmad Khonsari
- Investigating tip-selection optimization in IOTA/DAG-based distributed ledgers, with a current focus on mitigating parasite-chain attacks.
- Developed and evaluated a priority-based tip-selection mechanism using the official IOTA private Tangle and Hornet node software to reduce confirmation time for high-priority transactions.

L4S for Low-Latency Video Streaming ***2023 - 2024***
Research Project with ATHENA / University of Klagenfurt
- Investigated the use of Low Latency, Low Loss, Scalable Throughput (L4S) for low-latency video-streaming scenarios.
- Set up and experimented with a Mininet-based environment to study L4S behavior and low-latency transport mechanisms.

QoE Modeling in Volumetric Video Streaming ***2025 - 2026***
Research Project with Klagenfurt/ATHENA and IMDEA Networks
- Contributing to an ongoing survey on end-to-end Quality of Experience (QoE) modeling for volumetric video streaming.
- Responsible for reviewing and synthesizing subjective QoE assessment studies, including assessment methodologies, interaction settings, streaming impairments, and dataset availability.

TEACHING & ACADEMIC LEADERSHIP

University of Tehran ***2024 - 2025***

Chief Teaching Assistant / Teaching Assistant

- Chief TA - Distributed Systems: led a 6-member TA team for approximately 90 undergraduate students; coordinated project and exam design, grading, and course logistics.
- Chief TA – Blockchain Theory: contributed to exam design, grading, student support and technical troubleshooting.
- TA - Network Laboratory (two semesters): independently ran one weekly lab section, evaluated experiments, and supported students during hands-on networking exercises.
- TA - Internet Measurement: assisted with course assignments and student support.
- TA - Design Validation for HDL Models: assisted with project design, project evaluation, and student support.

RESEARCH MENTORING

Undergraduate Research Mentor ***2024 - Present***

University of Tehran

- Mentored three undergraduate research projects, from scoping and technical onboarding through implementation and evaluation.
- Projects included collecting data from the IOTA network, RL-based tip selection for parasite-chain detection, and distributed training systems research.
- Guided one student from initial distributed-learning concepts to implementation and manuscript contribution as the second author on the ongoing TCC project.

TECHNICAL & LABORATORY INFRASTRUCTURE

Network Laboratory Infrastructure Project

University of Tehran

- Installed VMware ESXi on a new laboratory server and integrated hosts through vSphere for centralized virtual-machine management.
- Deployed Prometheus-based server monitoring and configured Grafana dashboards for resource and system metrics.
- Deployed and administered PNETLab, created course sections and environments, and trained students in its use.
- Migrated and adapted laboratory exercises from GNS3 to PNETLab and consolidated the revised experiments into a unified lab manual.

ENTREPRENEURSHIP & OUTREACH

Co-Founder & Product Manager

Cryptocurrency Exchange Startup

- Co-founded a cryptocurrency exchange startup and led product planning, competitor analysis, feature comparison, and product prioritization.
- Delivered public workshops for roughly 50-100 participants and university seminars covering blockchain fundamentals, on-chain concepts, and digital-asset investment basics.

SELECTED TECHNICAL PROJECTS

Blockchain Node & Testnet Deployment

Graduate course project - University of Tehran

- Implemented a blockchain node, deployed it on a test network, and generated a test transaction while studying Bitcoin-style transaction and node workflows.

Internet Measurement & Monitoring

Graduate course project - University of Tehran

- Deployed Elasticsearch for system monitoring and configured Grafana components for metric visualization and operational inspection.

Independent Distributed Machine Learning Study

Ongoing self-study

- Completed MPI-based distributed-computing exercises and currently working through hands-on distributed and federated learning implementations.

TECHNICAL SKILLS

Systems & Infrastructure: Linux, Docker, Docker Compose, VMware ESXi, vSphere, PNETLab, Mininet, Git/GitHub
Programming: Python, C
Monitoring / Experimentation: Elasticsearch, Grafana, IOTA/Hornet; experience adapting and extending research simulators